

# Classification of images Project

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
```

```
data_train_path = 'Fruits_Vegetables/train'
data_test_path = 'Fruits_Vegetables/test'
data_val_path = 'Fruits_Vegetables/validation'
```

```
img_width = 180
img_height = 180
```

```
data_train=tf.keras.utils.image_dataset_from_directory
            (data_train_path,shuffle=True,
             image_size=(img_width,img_height),batch_size=32,
             validation_split=False)
```

```
data_train.class_names
```

```
data_cat = data_train.class_names
```

[illegible][illegible]

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```
plt.figure(figsize=(10,10))

for image,labels in data_train.take(1):

    for i in range(9):
        plt.subplot(3,3,i+1)
        plt.imshow(image[i].numpy().astype('uint8'))
        plt.title(data_cat[labels[i]])
        plt.axis('off')
```

```
from tensorflow.keras.models import Sequential
```

```
model = Sequential([
    layers.Rescaling(1./255),
    layers.Conv2D(16, 3, padding='same', activation='relu'),
    layers.MaxPooling2D(),
    layers.Conv2D(32,3, padding='same',activation='relu'),
    layers.MaxPooling2D(),
    layers.Conv2D(64, 3, padding='same', activation='relu'),
    layers.MaxPooling2D(),
    layers.Flatten(),
    layers.Dropout(0.2),
    layers.Dense(128),
    layers.Dense(len(data_cat))
])
```

```
model.compile(optimizer='adam',
loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=True),
metrics=['accuracy'])
```

```
epochs_size = 25
history = model.fit(data_train, validation_data=data_val, epochs=epochs_size)
```

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```
epochs_range = range(epochs_size)
plt.figure(figsize=(8,8))
plt.subplot(1,2,1)
plt.plot(epochs_range,history.history['accuracy'],label = 'Training Accuracy')
plt.plot(epochs_range, history.history['val_accuracy'],label = 'Validation Accuracy')
plt.title('Accuracy')

plt.subplot(1,2,2)
plt.plot(epochs_range,history.history['loss'],label = 'Training Loss')
plt.plot(epochs_range, history.history['val_loss'],label = 'Validation Loss')
plt.title('Loss')
```

```
image = 'banana.jpg'
image = tf.keras.utils.load_img(image, target_size=(img_height,img_width))
img_arr = tf.keras.utils.array_to_img(image)
img_batch=tf.expand_dims(img_arr,0)
```

```
predict = model.predict(img_batch)
```

```
score = tf.nn.softmax(predict)
```

```
print('Veg/Fruit in image is {} with accuracy of {:.2f}'.format(data_cat[np.argmax(score)],np.max(score)*100))
```

```
model.save('Image_classify.keras')
```

## App Deployment

```
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras.models import load_model
import streamlit as st
import numpy as np

st.header('Image Classification Model')

model = load_model(r'C:\Users\1\Desktop\deep
learn\Fruits_Vegetables\Image_classify.keras')
data_cat = ['apple',
            'banana', 'beetroot', 'bell pepper', 'cabbage', 'capsicum', 'carrot',
            'cauliflower', 'chilli pepper', 'corn', 'cucumber', 'eggplant',
            'garlic', 'ginger', 'grapes', 'jalepeno', 'kiwi', 'lemon', 'lettuce',
            'mango', 'onion', 'orange', 'paprika', 'pear', 'peas', 'pineapple',
            'pomegranate', 'potato', 'raddish', 'soy beans', 'spinach',
            'sweetcorn', 'sweetpotato', 'tomato', 'turnip', 'watermelon']

img_height = 180
img_width = 180
image = st.text_input('Enter Image name', 'Apple.jpg')

image_load = tf.keras.utils.load_img(image, target_size=(img_height, img_width))
img_arr = tf.keras.utils.array_to_img(image_load)
img_bat = tf.expand_dims(img_arr, 0)

predict = model.predict(img_bat)

score = tf.nn.softmax(predict)
st.image(image, width=200)
st.write('Veg/Fruit in image is ' + data_cat[np.argmax(score)])
st.write('With accuracy of ' + str(np.max(score)*100))
```